

## EMPATHIZE & DISCOVER

### 1. Understanding the User

The Empathize and Discover phase focuses on understanding the problems, expectations, frustrations, and requirements of students who need quick access to academic information.

The primary users identified for the system are **Students / Users** and **Administrators**. The student is the main user who interacts with the FAQ and AI Assistant features.

### 2. User Persona

| User Type         | Student                                 |
|-------------------|-----------------------------------------|
| Primary Goal      | Get quick and relevant academic answers |
| Main Need         | Easy access to FAQ information          |
| Main Problem      | Difficulty finding the required FAQ     |
| Expectation       | Fast and relevant response              |
| Technology Used   | Web browser / Computer / Mobile         |
| Major Frustration | Searching through multiple FAQ entries  |

### 3. Empathy Map

| "I need the answer quickly."<br>"Where can I find this information?"<br>"There are too many FAQs." | "Will the system understand my question?"<br>"Is this the correct answer?"<br>"Can I find related questions?" | Academic instructions<br>Questions from classmates<br>Course-related information<br>Examination discussions | Confused when information is difficult to find<br>Frustrated by repeated searching<br>Satisfied when answers are found quickly |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|

### 4. User Pain Points

| Manual Searching           | Users may need to browse multiple FAQ entries before finding the required information.          |
|----------------------------|-------------------------------------------------------------------------------------------------|
| Different Question Wording | The same question can be expressed using different words or sentence structures.                |
| Information Overload       | A large FAQ collection can make it difficult to identify the relevant answer.                   |
| Repeated Questions         | Students may repeatedly ask the same basic questions when information is not easily accessible. |

### 5. User Gains

- Faster information retrieval.
- Simple question-answer interaction.
- Relevant FAQ matching.
- Related FAQ suggestions.
- Organized FAQ categories.
- Easy web-based access.
- Centralized academic information.

## 6. User Needs

1. Simple and user-friendly interface.
2. Secure authentication.
3. Question-based interaction.
4. Relevant FAQ matching.
5. Category-based FAQ browsing.
6. Related FAQ suggestions.
7. Easy admin management.
8. Clear and understandable responses.

## 7. Opportunity Identified

The traditional FAQ system can be improved by introducing an intelligent matching mechanism. Instead of requiring users to search for an exact question, the AI FAQ Assistant processes the user's input and compares it with stored FAQ questions, keywords, and answers to identify the most relevant information.